Cameroon Food Price Watch

A monthly pipeline that flags unusual food prices, and tells a one-off blip from a real change in the market

Bih Brandy Sango · Data science and engineering · November 2026

Early warning for unusual food prices

1

The problem

Prices swing with the seasons, so families, traders and NGOs can't tell if today's price is normal until it's too late.

2

What I did

Built a pipeline for 76 markets that cleans and tests prices, flags the unusual ones, says why, and forecasts next month.

3

What changed

It caught a maize crash in 15 of 18 markets, found errors in WFP's own data, and runs by itself every month.

The data: World Food Programme prices

68,996
prices, January 2017 to September 2026

76
markets across 9 regions

54
foods, retail and wholesale

29
markets still reporting in September 2026

Read through the official API of the Humanitarian Data Exchange (CC BY-IGO). Coverage is shrinking: Limbé, Tiko, Mutengene and Muea stopped reporting after December 2025.

Method: five steps, run every month

1

Fetch

Python, from the HDX API, 2,000 rows at a time; re-checks the last six months.

2

Clean and store

dbt and DuckDB; every bag size turned into a price per kg; suspect prices flagged.

3

Compare with normal

Same market, food and month in the previous five years only. 20% off: watch. 40%: alert.

4

Say why

General shift, market-wide shift, local and lasting, or local and new.

5

Forecast and report

A tested forecast and a seven-page Power BI report. 21 tests and 27 checks.

Decisions, and why I made them

The official API, not scraping

The download link blocks scripts, so I used the supported route instead of working around it.

Median of earlier years only

One extreme year can't distort normal, and an alert never uses data from later on.

Flag bad prices, don't delete them

1,052 suspect prices stay visible for checking but never trigger an alert.

A range, not one number

The forecast isn't more accurate than a simple guess, so the report shows a likely range.

Maize is in a price crash

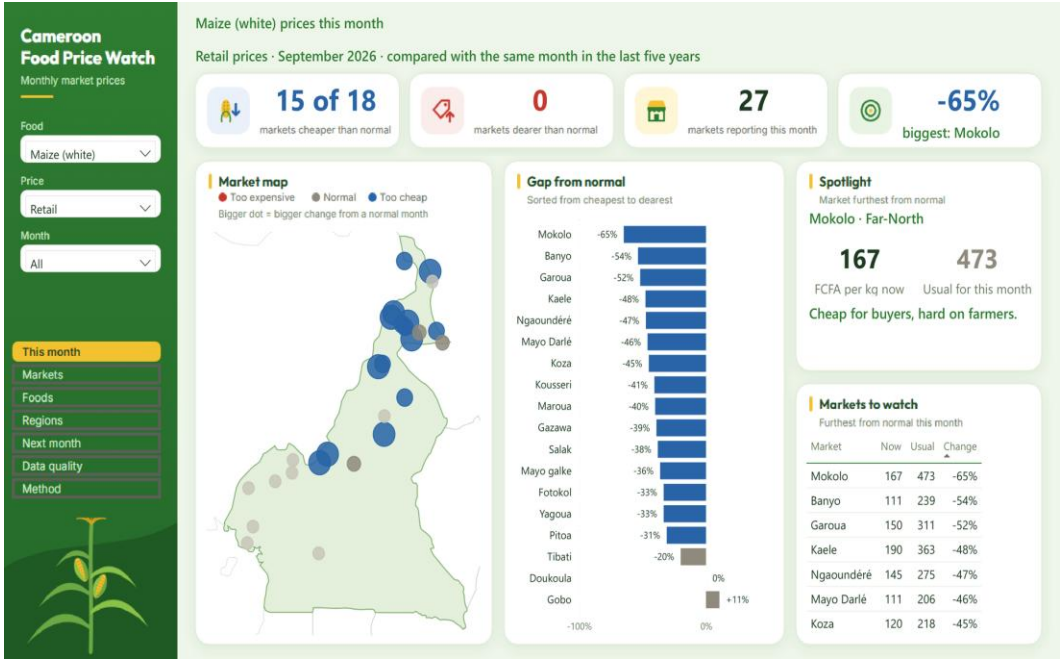
15 of 18

markets sold maize well below normal in September 2026

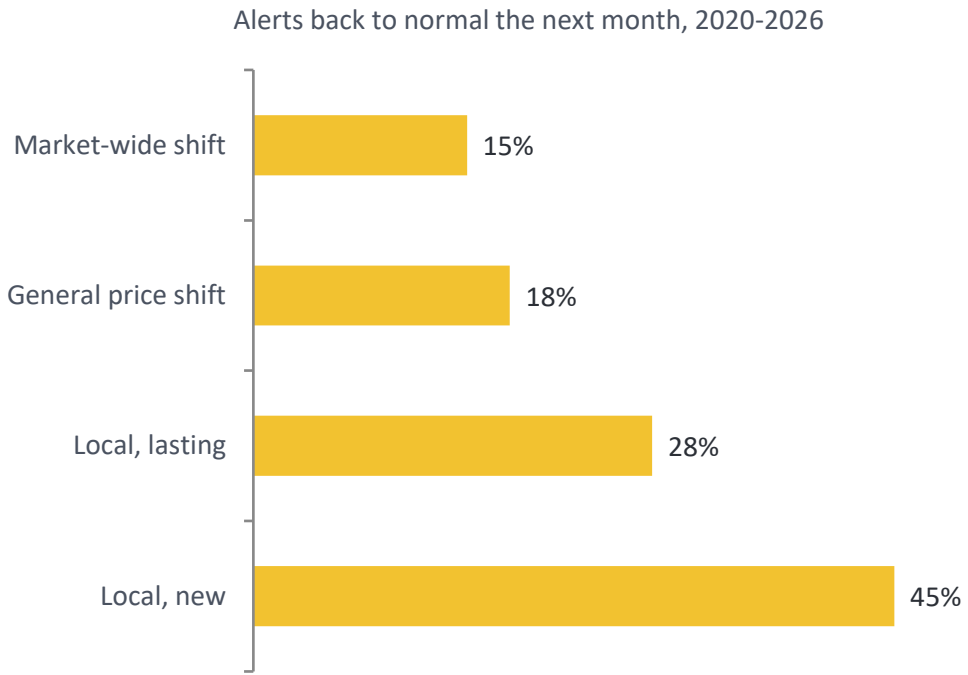
−65%

Mokolo, below its usual September price (Garoua −52%, Maroua −40%)

Good for families who buy maize, hard for farmers.



Blip or real change? Alert types tell them apart



3×

A lone, first-time alert vanishes three times as often as a market-wide one, so it deserves a check before anyone acts.

General price shifts appear only in 2020-2024, peaking in 2023 when food inflation was high.

An honest forecast, and errors the checks caught

3.3%

typical forecast error ("same as last month": 3.2%)

6 in 10

direction right when a price really moved

35% of prices don't change from month to month, so "no change" is hard to beat. The report shows a range, not one price.

Errors found in the source

- September 2023: many markets jump 3 to 10 times, then drop back
- April 2024: cassava bag prices entered as per kg
- 1,052 prices (1.5%) flagged and kept out of alerts

What it can't tell you, and what I'd do next

Limits

- "Normal" isn't adjusted for official inflation yet
- 78% of prices can't be judged: markets need three years of history
- The suspect check misses errors that hit every market at once
- One month ahead is a short horizon

Next steps

- Use Cameroon's consumer price index
- Compare each price with the same market's previous month
- Test rainfall and exchange-rate data in the forecast

Public data can warn people early about unusual food prices, as long as the system is honest about what's normal, what's a blip and what its forecast can do.

Thank you. Questions?

Project page: bihbrandysango-cloud.github.io/projects/food-price-watch
Code: github.com/bihbrandysango-cloud/cameroon-food-price-watch

Backup: questions you may ask

- Why didn't your forecast beat the simple guess?
- How do you know an alert is real and not a mistake?
- Why the median of five years?
- What would you do with more time?
- Who is this for?
- What was the hardest part?
- How do you make sure it keeps working?